Vid2Slide
User Documentation

Version 1.0.0

EPR Groupers

May 2024

Contents

1. Introduction

1.1 What is Vid2Slide?

Vid2Slide is a powerful desktop application designed to intelligently extract unique slides from lecture videos, presentations, and recorded sessions. Using advanced computer vision algorithms, it automates the tedious process of manually extracting frames from hours of video content.

1.2 Key Capabilities

- **Automatic Deduplication**: Uses perceptual hashing to detect and remove duplicate slides
- **Blur Detection**: Filters out blurry transition frames automatically
- **Gallery Preview**: Review and edit extracted slides before export
- **Multi-Format Export**: Export to PDF, ZIP, or PowerPoint
- **Fully Offline**: No internet required, complete privacy

1.3 System Requirements

Table 1: Minimum and Recommended System Requirements

Component	Minimum	Recommended
Operating System	Windows 10 (64-bit)	Windows 11
Processor	Dual-core 2.0 GHz	Quad-core 3.0 GHz
RAM	4 GB	8 GB or more
Storage	500 MB free	2 GB free
Graphics	Integrated	NVIDIA GPU (CUDA)
Display	1280x720	1920x1080

2. Installation

2.1 Download Options

Installer Version (Recommended)

1. Visit https://github.com/Supankan/Vid2Slide/releases
2. Download the latest `Vid2Slide-Setup.exe`
3. Run the installer and follow the wizard

Portable Version

1. Download the `.zip` release package
2. Extract to any folder of your choice
3. Run `Vid2Slide.exe` directly

2.2 Installation Steps

1. Launch the downloaded `Vid2Slide-Setup.exe`
2. Select your preferred installation directory
3. Choose whether to create a Desktop shortcut
4. Click **Install** to begin installation
5. Once complete, click **Launch** to start Vid2Slide

2.3 Post-Installation Setup

After first launch, Vid2Slide will:

- Create necessary folders in `%APPDATA%/Vid2Slide/`
- Check for updates (if enabled in settings)
- Display the main interface ready for use

2.4 Uninstallation

To uninstall Vid2Slide:

1. Go to **Settings** then **About**
2. Click **Uninstall**
3. Follow the uninstallation wizard

Note: User data and settings are preserved in `%APPDATA%/Vid2Slide/` unless you check **Remove user data**.

3. Getting Started

3.1 Interface Overview

The Vid2Slide interface consists of four main areas:

1. **Toolbar**: Import video, export, settings access
2. **Video Preview**: Displays current video with playback controls
3. **Processing Controls**: Settings and extraction buttons
4. **Slide Gallery**: Grid view of extracted slides

3.2 Importing a Video

Method 1: Button Import

1. Click the **Import Video** button in the toolbar
2. Browse and select your video file
3. The video will load in the preview area

Method 2: Drag and Drop

1. Drag any supported video file onto the application window
2. Drop to automatically import

3.3 Supported Video Formats

Table 2: Supported Video Formats

Format	Extension	Codecs Supported
MP4	.mp4	H.264, H.265/HEVC, VP9
Matroska	.mkv	All FFmpeg supported codecs
AVI	.avi	Various codecs
QuickTime	.mov	ProRes, H.264, HEVC
WebM	.webm	VP8, VP9, AV1

3.4 Video Playback Controls

Once a video is loaded, use these controls:

- **Play/Pause**: Toggle video playback
- **Seek Bar**: Scrub through the video
- **Volume**: Adjust audio level
- **Speed**: 0.5x, 1x, 1.5x, 2x playback
- **Frame Step**: Step forward/backward one frame

4. Processing Configuration

4.1 Understanding the Algorithms

Vid2Slide uses two primary algorithms for slide extraction:

Perceptual Hashing (pHash)

Unlike traditional hashing where a single pixel change creates a completely different hash, perceptual hashing produces similar hashes for visually similar images.

How it works:

1. Reduce image to a small size (32x32 pixels)
2. Apply Discrete Cosine Transform (DCT)
3. Keep only the low-frequency components
4. Convert to a binary hash string
5. Compare hashes using Hamming distance

Two frames with a Hamming distance 5% or less are considered duplicates.

Blur Detection (Laplacian Variance)

The Laplacian operator highlights edges in an image. Sharp images have high variance in edge detection, while blurry images have low variance.

Frames with Laplacian variance **below 100** are considered blurry and discarded.

4.2 Processing Settings

Table 3: Processing Settings

Setting	Description	Default	Range
Similarity Threshold	How similar frames must be to merge	0.95	0.85 - 0.99
Blur Threshold	Laplacian variance cutoff	100	50 - 200
Min Slide Duration	Minimum seconds a slide appears	1s	0.5s - 5s
Max Slides	Limit total slides (0 = unlimited)	0	0 - 500
Frame Sampling Rate	How often to sample frames	Auto	Manual/Auto

4.3 Threshold Adjustment Guide

For Lecture Videos (Presenter + Slides):

- Similarity: 0.92 - 0.95

- Blur: 80 - 120
- Rationale: Captures all slide transitions without missing similar-looking slides

For Screen Recordings:

- Similarity: 0.88 - 0.92
- Blur: 60 - 100
- Rationale: Screen content has more variation, requires lower threshold

For Presentation Recordings:

- Similarity: 0.93 - 0.97
- Blur: 100 - 150
- Rationale: Clean transitions between slides

4.4 GPU Acceleration

For users with NVIDIA GPUs, enable CUDA acceleration:

1. Open **Settings** then **Processing**
2. Enable **Use GPU Acceleration**
3. Restart Vid2Slide for changes to take effect

Table 4: GPU vs CPU Performance

Operation	**CPU**	**GPU**
pHash calculation	Yes	Yes (10x faster)
Blur detection	Yes	Yes (8x faster)
Image loading	Yes	Yes (3x faster)
PDF generation	Yes	No

5. Slide Extraction Process

5.1 Starting Extraction

1. Ensure your video is loaded
2. Configure processing settings if needed
3. Click the **Extract Slides** button
4. Processing will begin automatically

5.2 Progress Monitoring

During extraction, the interface displays:

- **Progress Bar**: Visual representation of completion
- **Current Frame**: Shows the frame being analyzed
- **Slides Detected**: Running count of unique slides
- **Time Remaining**: Estimated time to completion

Typical processing times:

- 10-minute video: 2-3 minutes
- 1-hour video: 15-20 minutes
- 3-hour video: 45-60 minutes

5.3 Automatic Features

During processing, Vid2Slide automatically:

- Skips intro/outro sequences (if detected)
- Adjusts sampling rate based on scene complexity
- Groups similar frames together
- Filters blurry transitions
- Identifies slide boundaries

5.4 Batch Processing

To process multiple videos:

1. Go to **File** then **Batch Mode**
2. Add multiple video files to the queue
3. Set common processing settings
4. Click **Start Batch**

Batch processing continues even if individual videos fail.

6. Gallery View and Editing

6.1 Gallery Interface

After extraction completes, the Gallery View displays all extracted slides in a grid layout.

Gallery Features:

- **Thumbnail Grid**: Visual overview of all slides
- **Slide Numbering**: Sequential numbering for reference
- **Quick Actions**: Hover to reveal delete/reorder buttons
- **Zoom Preview**: Click to view in full resolution

6.2 Deleting Slides

To remove unwanted slides:

1. Hover over the slide thumbnail
2. Click the **X** button in the top-right corner
3. Confirm deletion

Tips for deletion:

- Delete duplicate slides even if threshold did not catch them
- Remove any blurry frames the algorithm missed
- Eliminate irrelevant content (credits, watermarks)

6.3 Reordering Slides

To rearrange the order:

1. Click and hold the slide thumbnail
2. Drag to the desired position
3. Release to drop
4. Other slides will shift automatically

The final PDF will follow this order.

6.4 Zoom and Inspect

Click any slide to open the Zoom View:

- Pan around the full-resolution image
- Zoom in/out with scroll wheel
- Compare slides side-by-side
- Press **ESC** to close

7. Export Options

7.1 PDF Export

The most common export format.

1. Click **Export to PDF**
2. Choose output location and filename
3. Configure page settings
4. Click **Export**

PDF Settings:

Table 5: PDF Export Options

Option	Available Values
Page Size	A4, Letter, Legal, A3, Custom
Orientation	Auto, Portrait, Landscape
Quality	Low (500KB), Medium (1MB), High (2MB), Maximum
Margins	0mm - 30mm
Image Resolution	720p, 1080p, 2K, Original

7.2 ZIP Archive Export

Export individual slides as images.

1. Select **Export as ZIP**
2. Choose image format (JPG/PNG)
3. Set resolution
4. Click **Export**

The ZIP file includes:

- All slide images numbered sequentially
- `metadata.json` with slide information
- `manifest.txt` with extraction details

7.3 PowerPoint Export

Convert slides to PowerPoint presentation.

1. Select **Export as PowerPoint**
2. Choose slide layout template
3. Optionally include speaker notes
4. Click **Export**

Each slide becomes a separate PowerPoint slide with the image embedded.

7.4 Quality Presets Comparison

Table 6: Export Quality Presets

Preset	Resolution	Size (10 slides)	Use Case
Draft	720p (1280x720)	500 KB	Quick preview, sharing
Standard	1080p (1920x1080)	2 MB	General use
High	2K (2560x1440)	5 MB	Print quality
Maximum	Original resolution	20 MB+	Archival, professional

8. Settings Reference

8.1 General Settings

- **Theme**: Light / Dark / System (default: Dark)
- **Language**: English, Spanish, French, German, Chinese, Japanese
- **Start Minimized**: Launch in system tray (default: Off)
- **Check for Updates**: Automatically download updates (default: On)

8.2 Export Settings

- **Default Page Size**: Preferred page size for PDF export
- **Default Quality**: Default quality preset
- **File Naming Pattern**: Original name, Custom prefix, Date-based
- **Open After Export**: Automatically open exported files

8.3 Advanced Settings

- **Temp Folder Location**: Where temporary files are stored
- **Clear Cache on Exit**: Free up space when closing
- **Max Memory Usage**: Limit RAM usage (for low-end systems)
- **Debug Mode**: Enable detailed logging

8.4 Keyboard Shortcuts

Table 7: Keyboard Shortcuts

Shortcut	**Action**
Ctrl + O	Open video file
Ctrl + E	Export slides
Ctrl + ,	Open settings
Delete	Remove selected slide
Ctrl + A	Select all slides
Space	Play/Pause video
Left/Right Arrow	Step frame backward/forward
Escape	Close zoom view

9. Troubleshooting

9.1 Common Issues

Video Won't Load

1. Verify the file is not corrupted (try opening in VLC)
2. Update video codecs (install K-Lite Codec Pack)
3. Check file read permissions
4. Ensure sufficient disk space (about 2x video size)

Processing is Very Slow

1. Enable GPU acceleration in settings
2. Close other applications to free RAM
3. Move video to SSD for faster read speeds
4. Reduce "Max Slides" limit

Export PDF is Blurry

1. Use "Maximum" quality preset
2. Increase resolution in export settings
3. Lower blur threshold if source is low quality

Application Crashes

1. Update to the latest version
2. Delete settings file to reset: `%APPDATA%/Vid2Slide/config.json`
3. Disable GPU acceleration temporarily
4. Run as Administrator

Slides are Duplicated

1. Lower similarity threshold to 0.92 or 0.90
2. Manually delete duplicates in Gallery View
3. Check for distinct visual styles in content

9.2 Error Codes

Table 8: Error Codes and Solutions

Code	Description	Solution
E1001	Video file not found	Re-import the video file
E1002	Unsupported codec	Install missing codec pack
E1003	Corrupted file	Re-download or re-encode video
E2001	Export path not writable	Choose different output location
E2002	Disk full	Free up space on drive
E3001	GPU not supported	Use CPU processing instead
E3002	Memory exceeded	Reduce batch size or restart app

9.3 Log Files

For detailed debugging, check log files:

1. Navigate to `%APPDATA%/Vid2Slide/logs/`
2. Open the most recent log file
3. Share relevant portions when reporting issues

To enable debug logging:

1. Go to Settings then Advanced
2. Enable "Debug Mode"
3. Restart Vid2Slide
4. Reproduce the issue
5. Check logs for detailed error information

10. Frequently Asked Questions

10.1 General Questions

Q: Why are some slides missing? A: Check your Blur Threshold setting. If set too high, legitimate slides may be filtered. Try lowering it to 50-80.

Q: Can I process multiple videos at once? A: Yes, use Batch Mode from the File menu to queue multiple videos for sequential processing.

Q: Does Vid2Slide work with screen recordings? A: Yes, but adjust Similarity Threshold to 0.90 or lower for screen content with more visual variation.

Q: My video is very long (more than 3 hours). Will it take longer? A: Processing time scales linearly. A 3-hour video takes approximately 15-30 minutes depending on hardware.

Q: Can I use Vid2Slide offline? A: Absolutely! Vid2Slide is fully offline. No internet required for processing or export.

Q: Does it work with non-English content? A: Yes, Vid2Slide is language-agnostic. It analyzes visual content, not text or audio.

10.2 Technical Questions

Q: What is perceptual hashing? A: A technique that generates similar hashes for visually similar images, allowing detection of duplicate slides even when they are not pixel-perfect matches.

Q: How accurate is blur detection? A: Blur detection uses Laplacian variance with a threshold of 100 by default. This catches most transition blurs but may miss very short blurry sections.

Q: What is the maximum video length? A: There is no hard limit, but very long videos (8+ hours) may require significant processing time and disk space. Consider splitting long videos.

Q: Can I process videos from a network drive? A: Yes, but processing will be slower. Copy videos to local drive for best performance.

10.3 Licensing Questions

Q: Is Vid2Slide free? A: Vid2Slide is free for personal and educational use. Commercial licensing is available upon request.

Q: Can I redistribute Vid2Slide? A: No, redistribution without permission is prohibited. Direct users to the official download page.

Q: Does the license cover academic institutions? A: Yes, educational institutions may use Vid2Slide under the same personal use terms. Contact us for enterprise licensing.

11. Support and Contact

11.1 Getting Help

For additional support:

- **GitHub Issues**: `https://github.com/Supankan/Vid2Slide/issues`
- **Email**: support@eprgroupers.com
- **Discord**: Join our community at `https://discord.gg/vid2slide`

11.2 Reporting Bugs

When reporting issues, please include:

1. Vid2Slide version (Help then About)
2. Windows version and build number
3. Video file format and duration
4. Error messages or screenshots
5. Log files from `%APPDATA%/Vid2Slide/logs/`

11.3 Feature Requests

We welcome feature suggestions! Submit your ideas at:

- **GitHub Discussions**: `https://github.com/Supankan/Vid2Slide/discussions`
- **Feedback Form**: Available in Help then Send Feedback

11.4 Contributing

Contributions are welcome:

- Submit pull requests for bug fixes
- Translate the interface to your language
- Create tutorials and documentation
- Report bugs and suggest features

12. Acknowledgments

Vid2Slide utilizes the following open-source libraries:

- **FFmpeg**: Video processing and codec support
- **OpenCV**: Computer vision and image processing
- **PDFtk**: PDF manipulation
- **Tailwind CSS**: Web interface styling

We are grateful to the open-source community for making these tools available.

Last updated: May 2024 | Version 1.0.0